Laksh Jain
Senior Undergraduate
Discipline of Artificial Intelligence
Indian Institute of Technology, Gandhinagar

1511.lakshjain@gmail.com
+91 6362588608
LinkedIn | Github | Google Scholar

ACADEMIC DETAILS

Degree	Specialization	Institute	Year	CPI/%
B.Tech.	*Artificial Intelligence*	IIT Gandhinagar	2023-27	9.05/10
Class XII	*Physics, Maths, Computer Sc.*	Narayana Junior College	2023	96.4 %
Class X		Sindhi High School	2021	95.0 %

ACHIEVEMENTS

- Published at **KDD 2026** in the *main track*, a leading ML **A* conference**. (*Project Website*)
- Received **Silver Medal** at **InterIIT Techmeet 14** for the ISRO Geospatial Problem Statement among **23** IITs.
- Awarded **Dean's List** and **Academic Citations** at IITGN for excellent academic performance.
- Secured an All-India Rank of **1906** in IIT JEE Advanced 2023 out of 1,89,744 candidates.
- Secured State Rank of **136** in Karnataka Common Entrance Test (KCET) 2023 out of 2,61,610 candidates.

PUBLICATIONS

- **ThermEval – A Structured Benchmark for Evaluation of Vision-Language Models on Thermal Imagery**
  Advisor: Prof. Nipun Batra, IIT Gandhinagar | ***KDD 2026 – Accepted, Poster (Datasets & Benchmarks Track)*** | *Paper*
  - Developed a large-scale benchmark of **70,000+** VQA samples to evaluate the reasoning capabilities of **25 open and closed sourced VLMs** (0.3B-235B parameters) under zero-shot and prompt-ablation settings. Built an end-to-end evaluation pipeline, including an **LLM-based output parser** to standardize responses across different model outputs.
  - Found that VLMs reliably identify thermal images as a modality but consistently **fail at temperature-grounded reasoning**, often defaulting to language priors (e.g., "normal" body temperature) instead of the actual thermal signal.
  - Showed the failure is a **grounding problem, not a scale problem** since errors persisted from 0.3B to 235B parameters.

WORK EXPERIENCE

- **AI Engineer Intern, Barclays** *May 2026 - July 2026*
  - Led the architecture and implementation of a GraphRAG pipeline on financial documents, integrating the system into the existing enterprise codebase while adhering to strict financial data governance and compliance frameworks.
  - Built **asynchronous REST APIs** using FastAPI on **Azure Functions** to serve LLM-backed tools for IB analysts, utilizing Azure AI Foundry, AI Search, Cosmos DB, and Blob Storage for reliable deployment at scale.
  - Implemented **binary quantization techniques** for vector embeddings within Azure AI Search, reducing vector index space utilization by up to **32x** while preserving retrieval speed and accuracy.
- **Summer Research Intern, IITGN** *May 2025 - July 2025*
  - Conducted research on zero-shot and fine-tuned evaluation of Vision-Language Models on thermal imagery; work culminated in ThermEval, accepted for poster presentation at **KDD 2026** (Datasets & Benchmarks Track).

PROJECTS

- **GeoNLI – Natural Language Interpretation of Satellite Imagery** *Nov 2025 – Dec 2025*
  (ISRO | Silver Medal @ Inter-IIT Tech Meet 14.0 @ IIT Patna) | *Project Link* | *Project Report*
  - Built an end-to-end neuro-symbolic pipeline for captioning, visual grounding, and VQA with geometric reasoning. **Fine-tuned** Qwen3-VL for caption generation and VQA, and integrated SAM3 for oriented object localization.
  - Achieved **2×** improvement over prior SOTA on VRSBench (**Acc@0.7 = 31.3%**) in grounding accuracy through a localize, refine and reason pipeline that disambiguates visually similar objects using spatial and attribute cues.
- **Layer Merging via Manifold Alignment for Model Compression** | Website | Github *Oct 2025 – Nov 2025*
  - Extended manifold-based layer merging (MKA) for LLM compression, reproducing **Llama3-8B** results with **43.75%** compression and only **2.8%** accuracy drop, while analyzing limitations in merge weights and layer similarity.
  - Experimented better merging strategies, including trainable merge weights (α), task and language specific layer similarity analysis, and neural alignment before merging to reduce performance loss caused by neuron misalignment.
  - Demonstrated that **alignment-aware merging** preserves accuracy under high compression, enabling stable merging of **13-15** layers with significantly **better accuracy retention** than naive averaging.
- **Supervised and Unsupervised Time Series Analysis on UCI-HAR Dataset** | Project Link *Aug 2024*
  - Analyzed the UCI-HAR dataset with time-series data of 30 subjects performing 6 activities. Applied **diffusion maps and K-Means**, achieving an ARI of 0.42 and a silhouette score of 0.68, outperforming PCA and t-SNE baselines.
  - Extracted features using the TSFEL library and trained a custom Decision Tree model. Validated it on real-world activity data collected via the Physics Toolbox Suite app, achieving **70% precision and 61% accuracy**.

TECHNICAL SKILLS

- **Languages**: Python, C, C++, SQL
- **Tools**: Git, GitHub, GitLab, VS Code, Azure (Blob Storage, Functions, Containers), Docker
- **Libraries/Frameworks**: PyTorch, HuggingFace, Pandas, FastAPI, Sklearn, PEFT, Weights Biases, Flask, Langgraph

RELEVANT COURSES

- **Completed Courses**: ML(**A**), Data Science(**A**), Maths for AI(**A**), NLP(**A-**), CV(**A-**), OS(**A-**), Databases(**A-**)